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Class: ECE 621
Date: 05/01/2022
Final Project

PROJECT TOPIC:

System Identification of a DC Servo Motor Using ARX and ARMAX Models

PAPER REFERENCE:

S. Mokhlis, S. Sadki and B. Bensassi, "System Identification of a DC Servo Motor Using ARX and ARMAX Models," 2019 International Conference on Systems of Collaboration Big Data, Internet of Things & Security (SysCoBIoTS), 2019

OBJECTIVE:

1. Generate data following the steps listed below:
    a. Build the physical system for generating PWM inputs to a servo motor and a potentiometer measuring the output angle of the servo.
    b. User selects from a list of pre-made signals or sends angles directly to the Arduino that then actuates the servo motor and records the input & output.
    c. Repeat the calculations using both model structures and varying input signals.
2. Using the generated data, create a model of the servo motor to make predictions of the servo motor's position using the input to the system.
3. Use prediction error method test the model against the system and analyze the results.

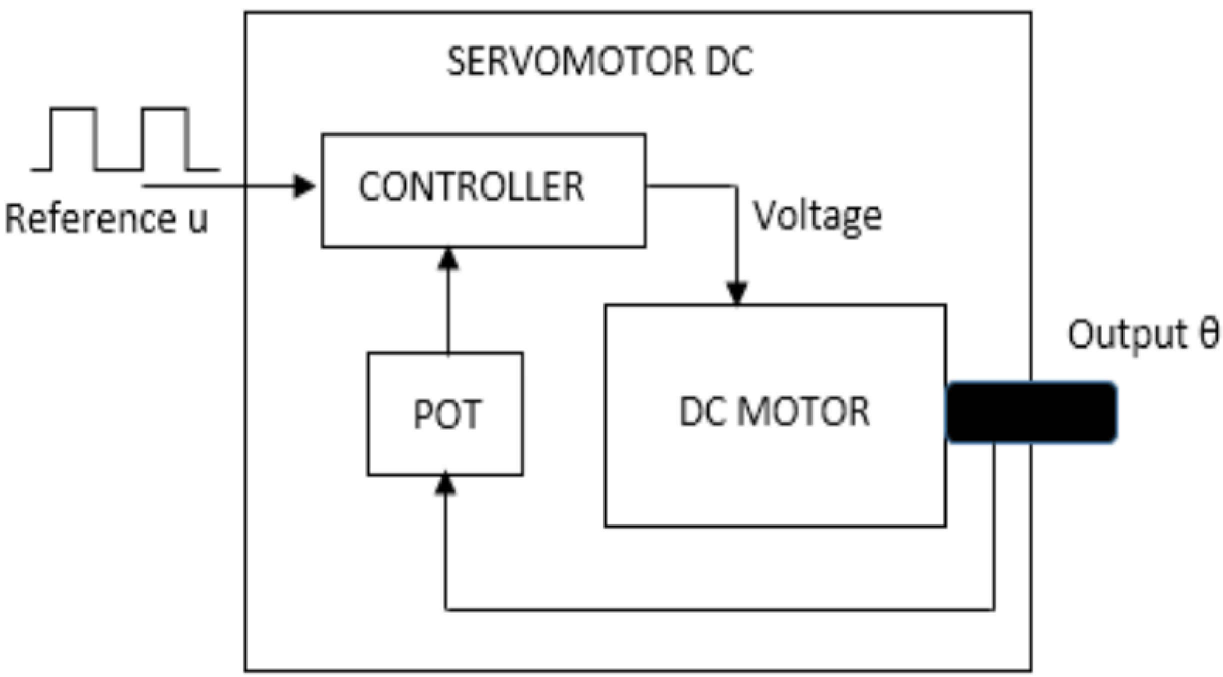
OVERALL APROACH:

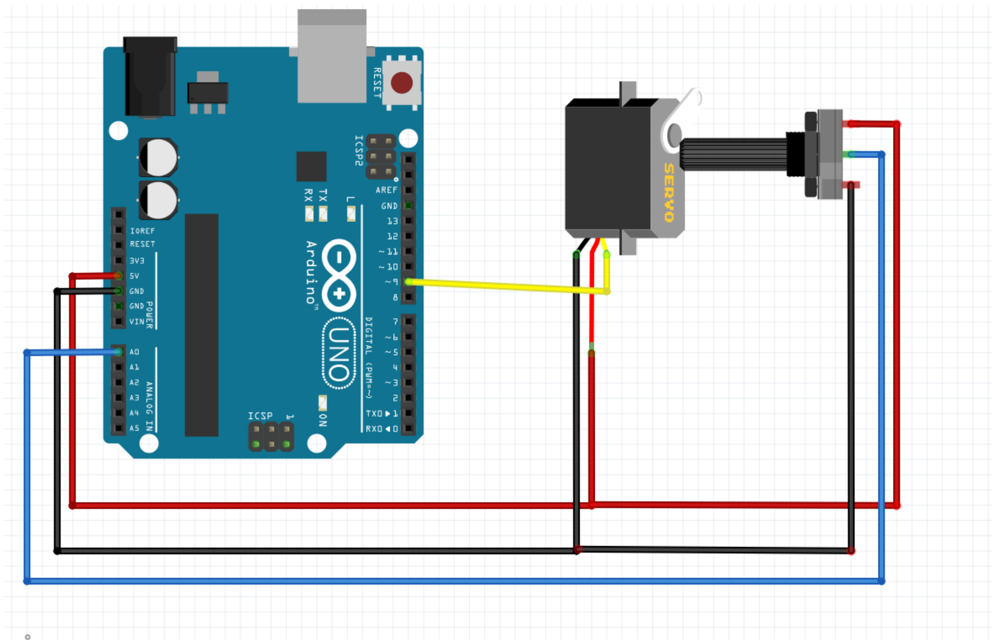
1. Operator selects an input signal to be used to model the output angle of the DC servo motor.
2. Angle predictor predicts how the DC servo output moves with respect to the input signal. The two models used for the algorithm are
    a. AutoRegressive with exogenous variable (ARX)
    b. AutoRegressive with adjusted Moving Average and exogenous variable (ARMAX)
3. The input signal is applied to the DC servo motor and its motion is logged by the potentiometer and microcontroller.
4. The model output and the system output are compared using the Prediction Error Method to refine the ARX and ARMAX models.

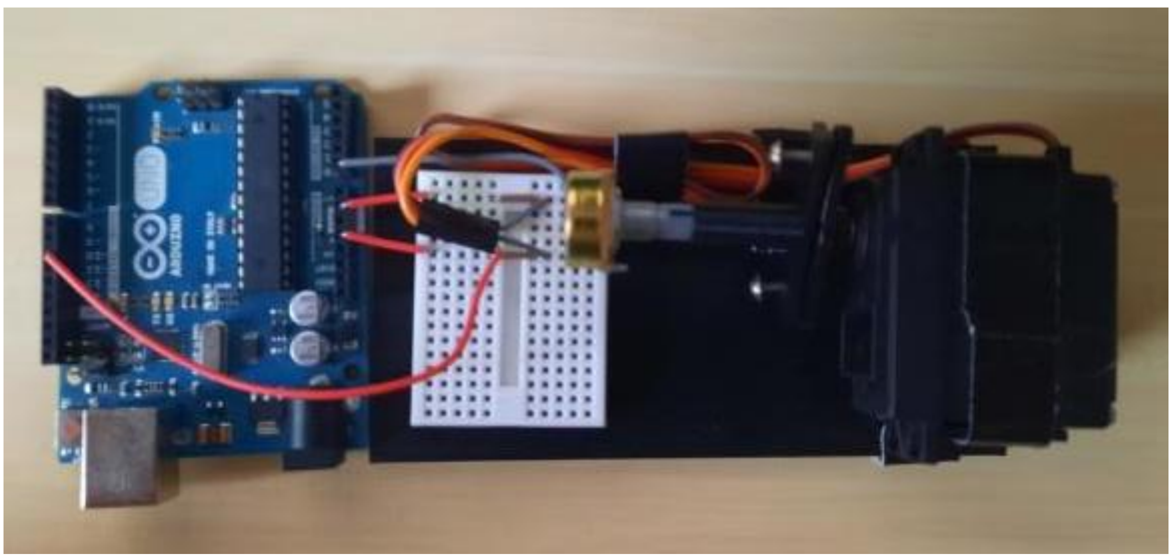
DATA COLLECTION:

A built a physical system utilizing a microcontroller, a DC servo motor, and a potentiometer to generated input signals to the DC servo motor. The system additionally measured the angle of the motor as it turned based on the generated PWM signals... The Arduino microcontroller then would log the input and output signals of the system to a .csv file. Multiple input scenarios were used to create models and the best models were compared against each other until a final one was chosen. These input signals included a step input, a square wave input, a sinusoid input and finally a random input.

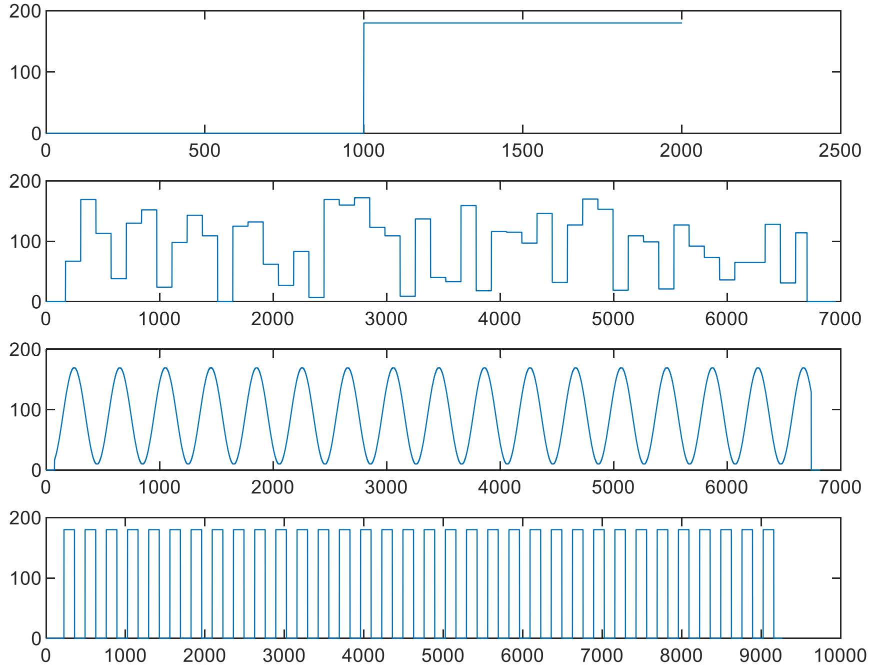
The system ran on 5V power that was provided from the Arduino. The yellow wire is the input PWM signal to the Arduino and the blue wire is the analog feedback from the potentiometer that was read in by the Arduino. Below is a block diagram, schematic, and picture of the system.



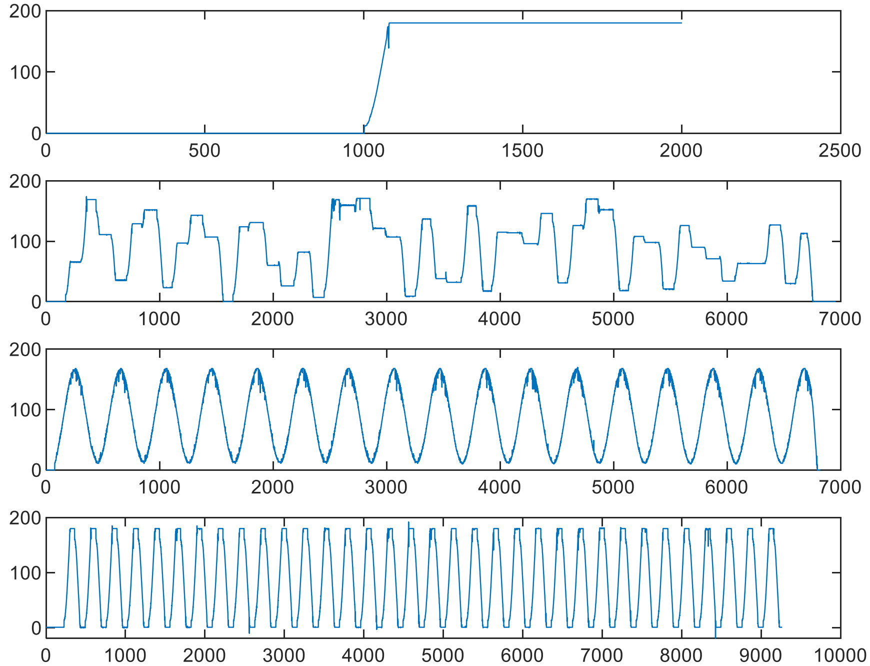




The data for this system was collected at approximately 60Hz. The servo's position can be tracked accurately to the degree and was given a set of different inputs to give necessary excitation of the system. Each one of these inputs can be seen in the graphs below.



Each one of these input signals resulted in the following output signals from the DC Servo motor system.



ARX ESTIMATION:

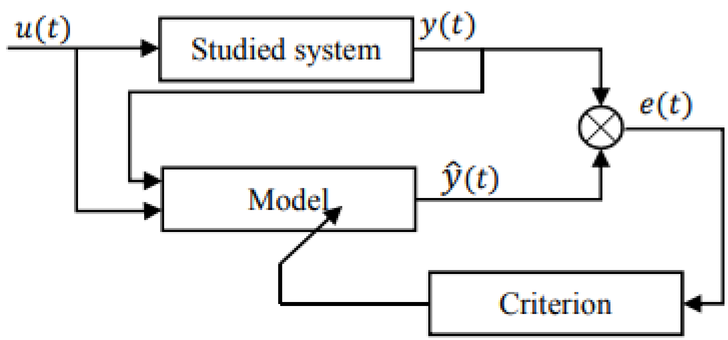
For this project, I focused on the two models that appeared in the paper. One of these models is the ARX model – AutoRegressive with an exogenous variable. This model can be defined by the equations.

$$Y_{arx}(k) = \frac{B(q)}{A(q)} q^{-nk} u(k) + \frac{1}{A(q)} \varepsilon(k)$$

$$A(q) = 1 + a_1 q^{-1} + \cdots + a_{n_a} q^{-n_a}$$

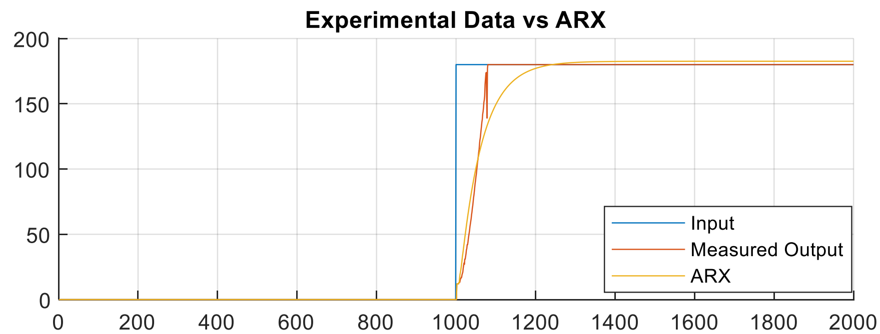
$$B(q) = b_1 q^{-1} + \cdots + a_{n_b} q^{-n_b}$$

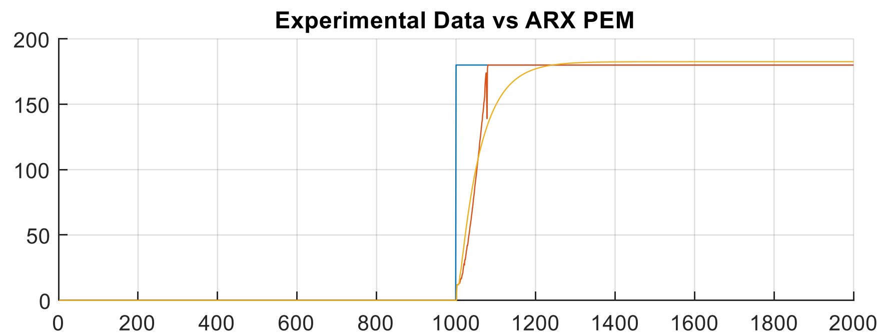
These functions constitute a transfer function that can define the system and its parameters. Additionally, the ARX estimation takes advantage of the Prediction Error Method to refine the estimation. PEM takes the input to the system and compares the predicted output and the measured output and will adjust the model's parameters to more accurately estimate the system. A diagram of the Prediction Error Method is shown below.



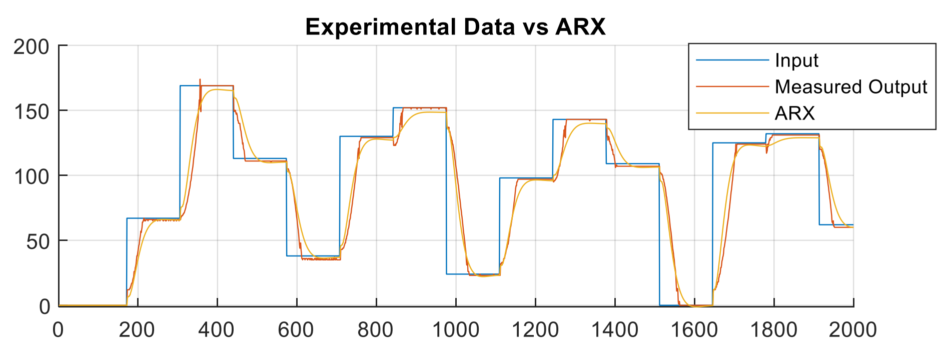
Using the tools provided in the MATLAB Systems Identification Toolbox, I built a model to estimate the parameters of the function and predict the output of the function. This was tested against all four input signals above, with the square wave signal yielding the most accurate model relative to the test data gathered. Below are graphs showing the four sets of data measured and the system estimation for that data set.

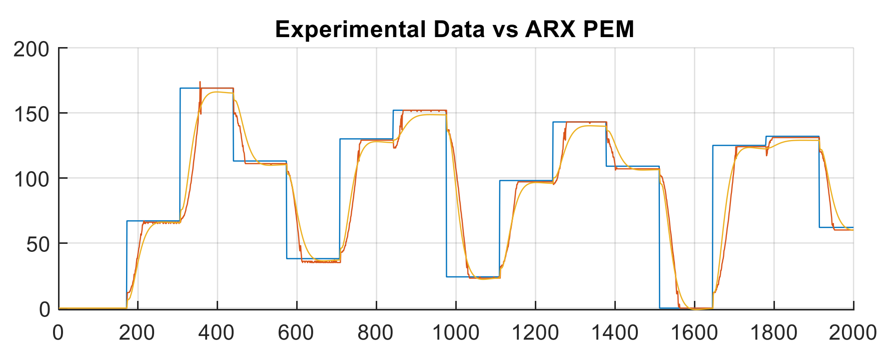
Step Input:



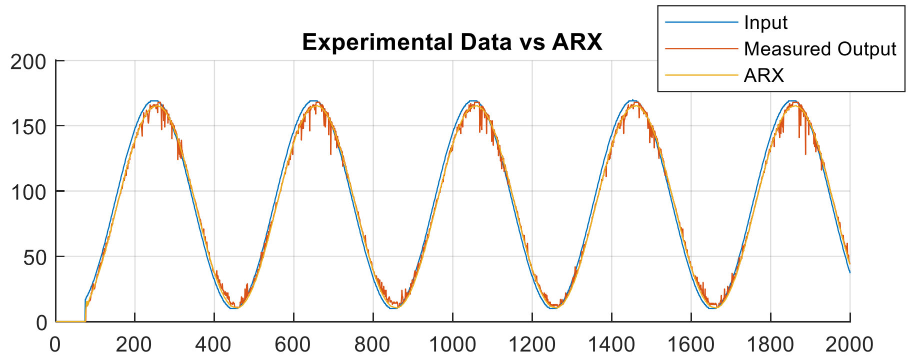


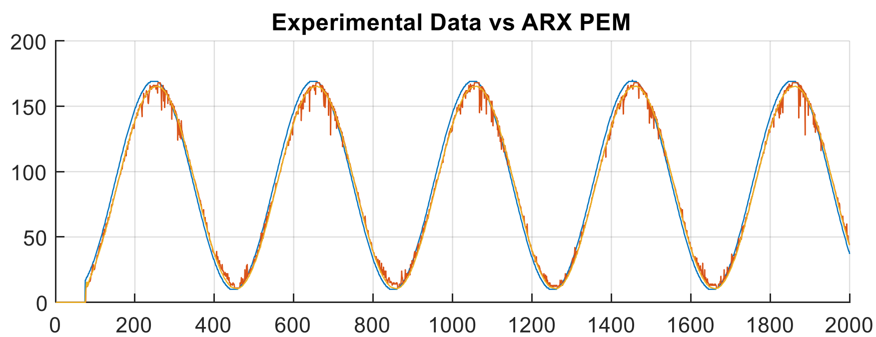
Randomly Generated Inputs:



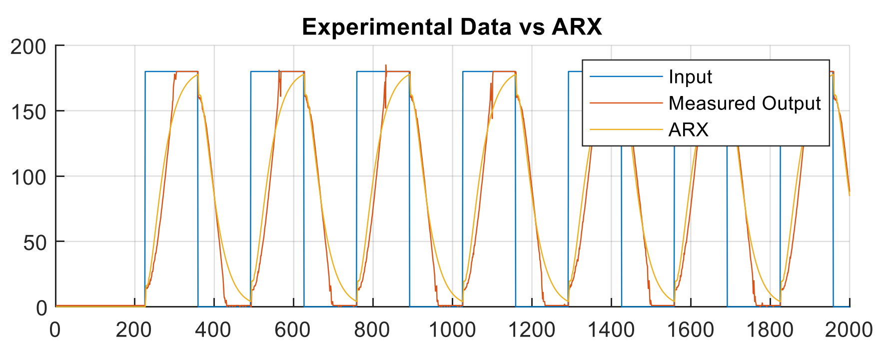


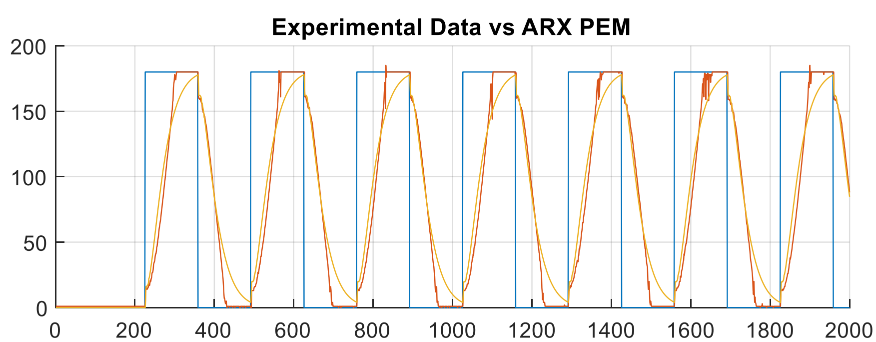
Sinusoid Input:





Square Wave Input:





Observations:

- Estimations for all four cases reach steady state estimation of parameters with accurate model predictions for all cases.
- Noise in the system doesn't appear to affect the estimated output.
- The ARX estimation tends to undershoot the estimations at the high and low points but is able to closely match the moving speed and holding behaviors of the servo motor.
- Each estimation's slight parameter changes offer large impacts on the system performance and prediction error.

ARMAX ESTIMATION:

The second model used in this project was the ARMAX Model – AutoRegressive Moving Average with exogenous variable. This model can be defined using the following equations.

$$Y_{armax}(k) = \frac{B(q)}{A(q)} q^{-nk} u(k) + \frac{C(q)}{A(q)} \varepsilon(k)$$

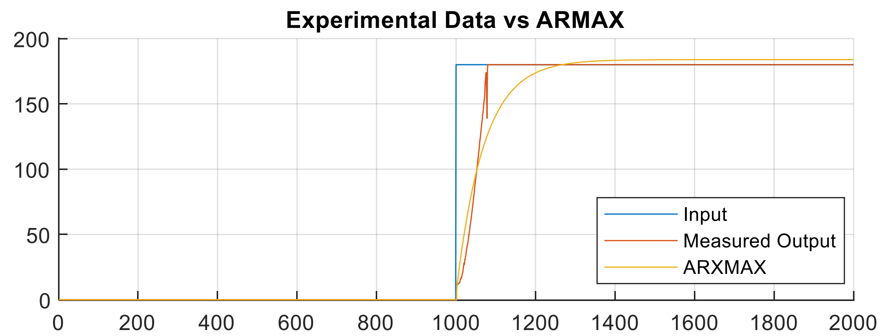
$$A(q) = 1 + a_1 q^{-1} + \cdots + \ a_{n_a} q^{-n_a}$$

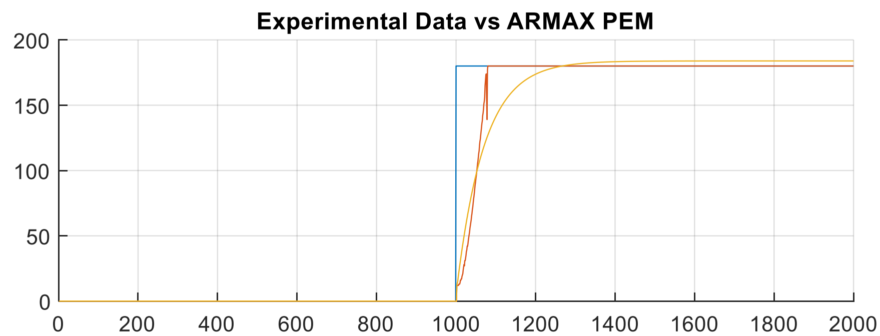
$$B(q) = b_1 q^{-1} + \cdots + \ a_{n_b} q^{-n_b}$$

$$C(q) = 1 + c_1 q^{-1} + \cdots + \ a_{n_c} q^{-n_c}$$

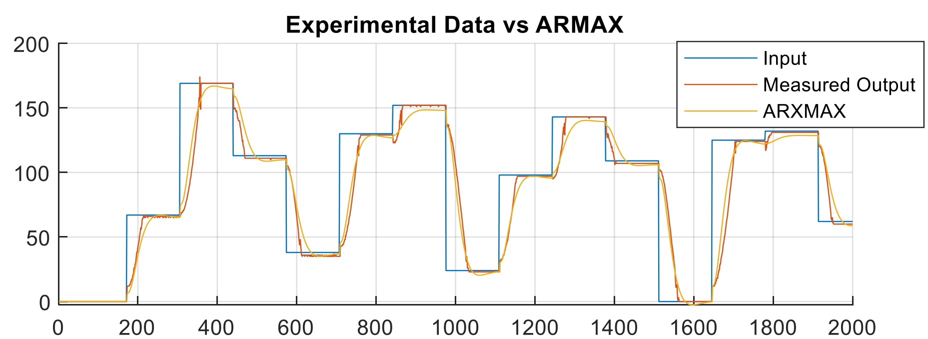
These functions define the transfer function for the system and by estimating the parameters of the system correctly, can allow the model to accurately predict the output of the studied system. The ARMAX estimation additionally takes advantage of the Prediction Error Method to refine its parameter estimation and allow for a better resultant model. Below are graphs of each input used for defining these models, the measured system output and the ARMAX model's output.

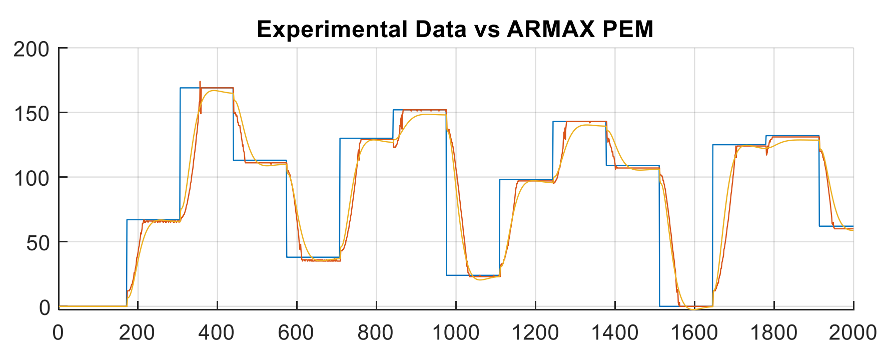
Step Input:



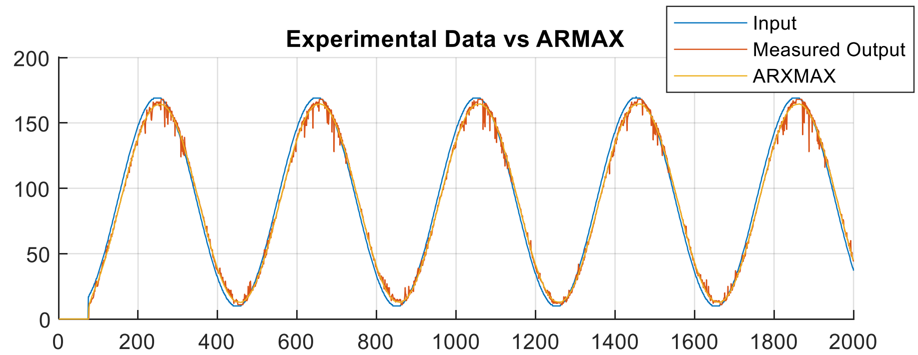


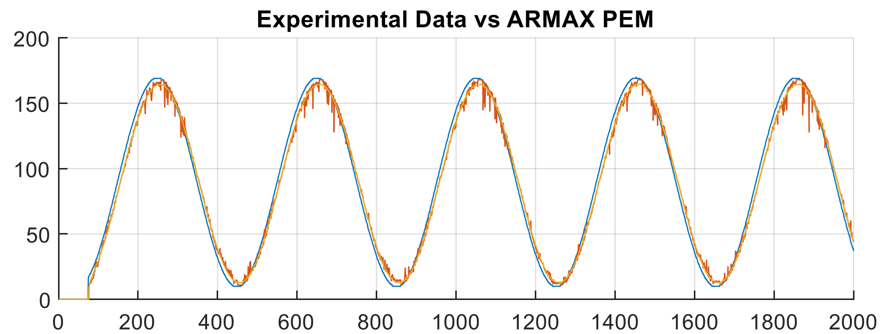
Random Input:



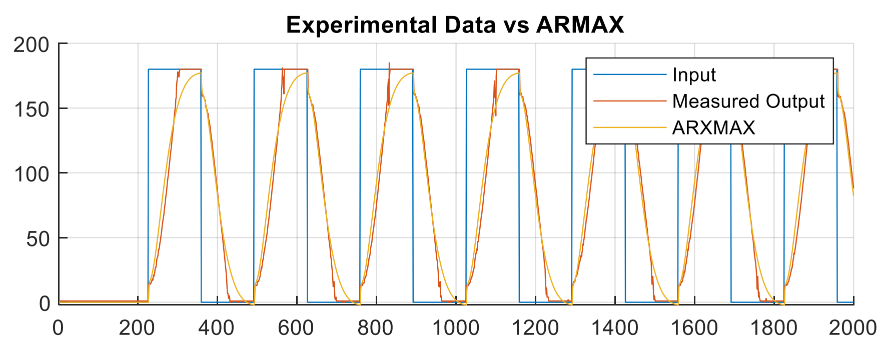


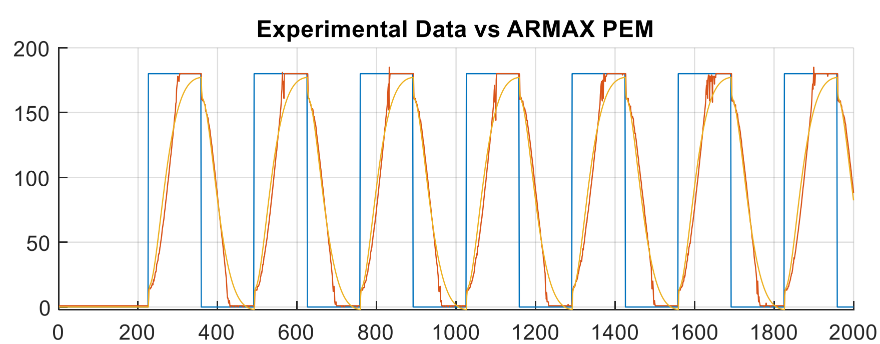
Sinusoid Input:





Square Wave Input:





Observations:

- Estimations for all four cases reach steady state estimation of parameters with accurate model predictions for all cases.
- Noise in the system doesn’t appear to affect the estimated output.
- The ARMAX estimation tends to overshoot the peak estimations but is able to closely match the other behaviors of the servo motor.
- Each estimation’s slight change in parameters yielded a much greater change in the ability of the system to accurately predict the movement of the motor, showing the importance of a input with sufficient excitation.

TEST DATA RESULTS:

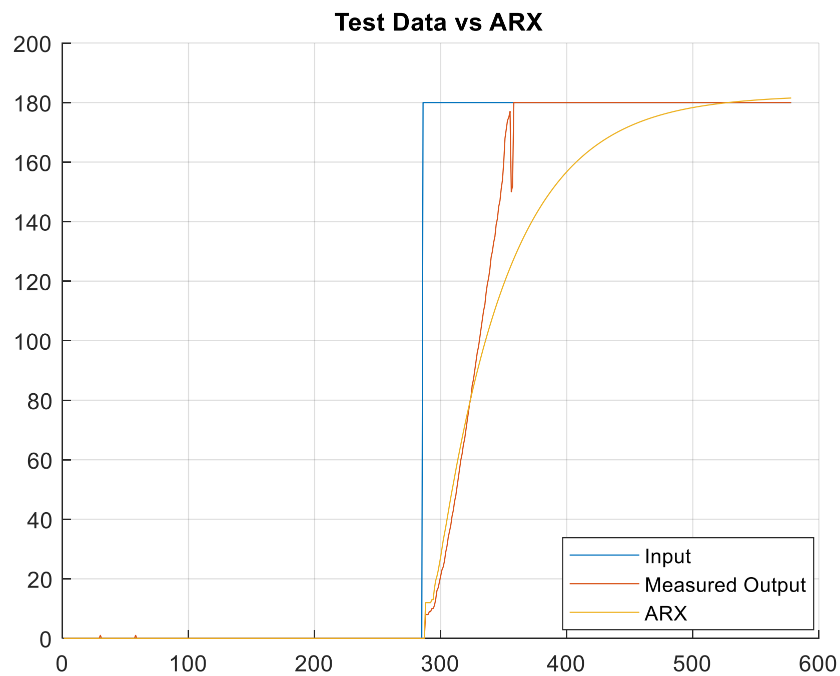
With multiple models created based on multiple input signals to the system, each created model was tested against test data. The test data was generated completely independently of the test input. The two cases tested against were random, aperiodic moves of the motor such as what could be expected in normal operation of the motor in a robotics format and a step input as a control case. For the following test results, only the models that have been adjusted with the prediction error method for more accurate estimates will be shown here. The FPE of each model was calculated using the ‘fpe’ function from the MATLAB system identification toolbox. Additionally, the best fit percentage was calculated by the ‘compare’ function from the system identification toolbox, using the normalized root mean square criterion which yields the same calculation used in the reference paper.

<u>**ARX Testing:**</u>

Step Input:

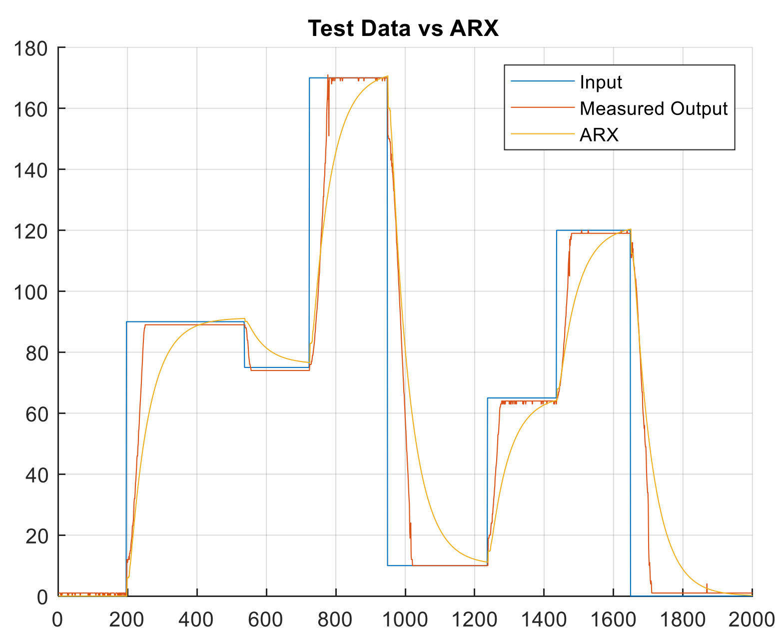
Best Fit %: 92.86%

FPE: 1.018

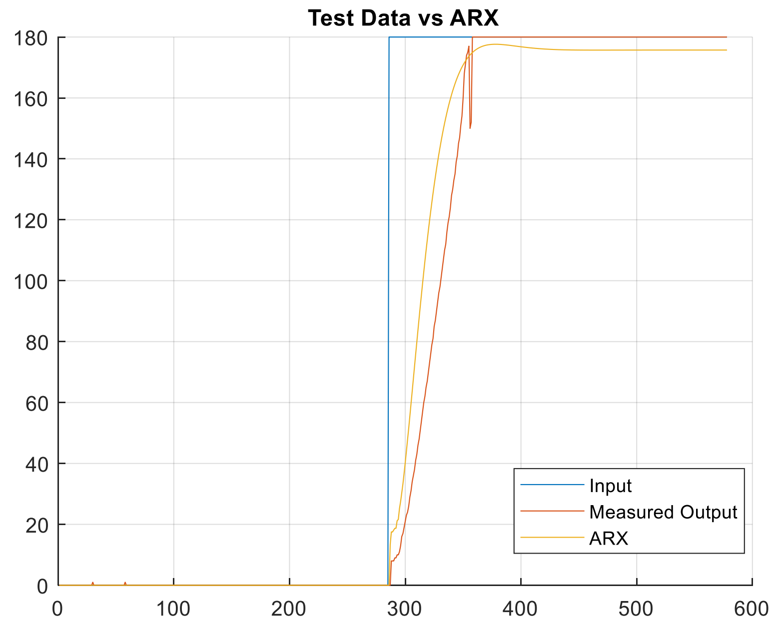


Best Fit %: 92.857%

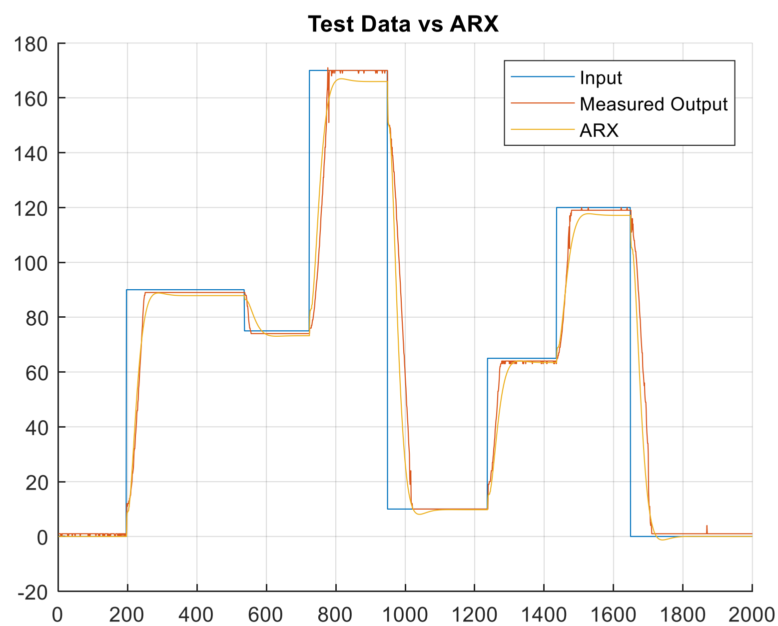
FPE: 1.0182



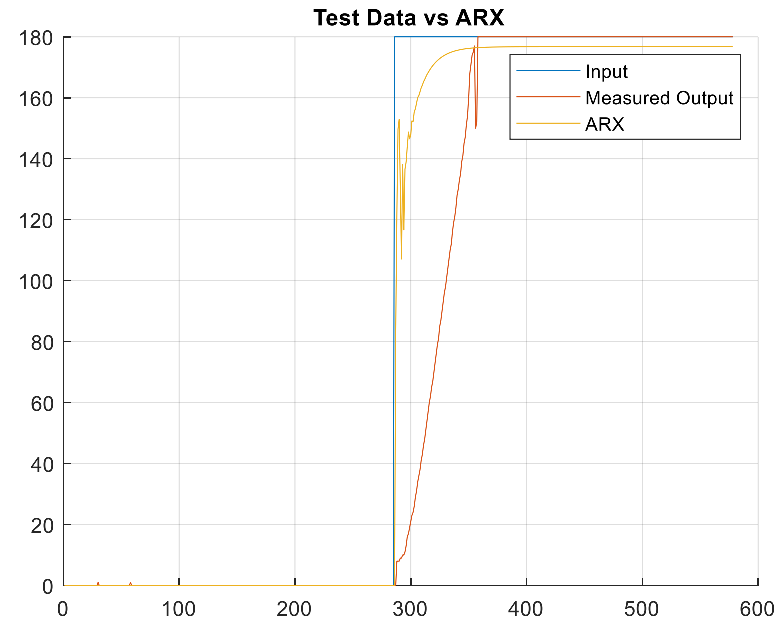
Random Input:
Best Fit %: 86.903%
FPE: 1.943



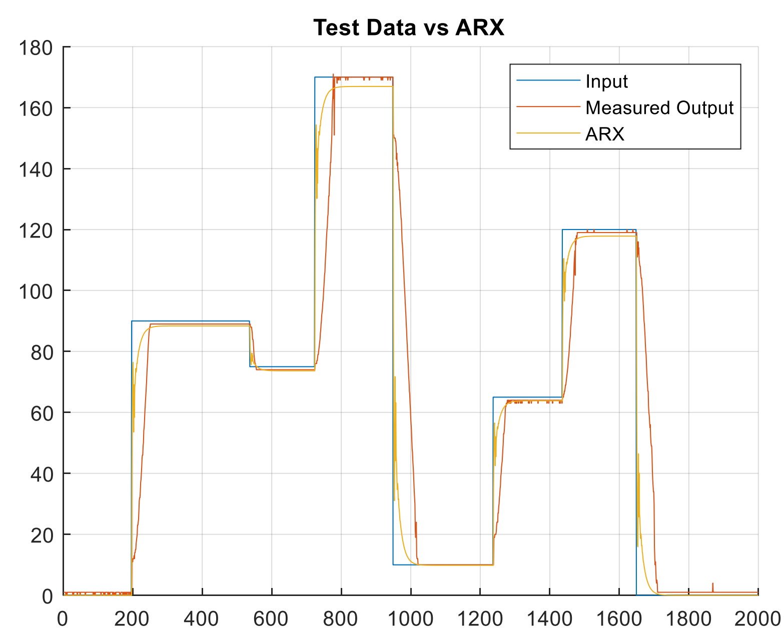
Best Fit %: 86.904%
FPE: 1.943



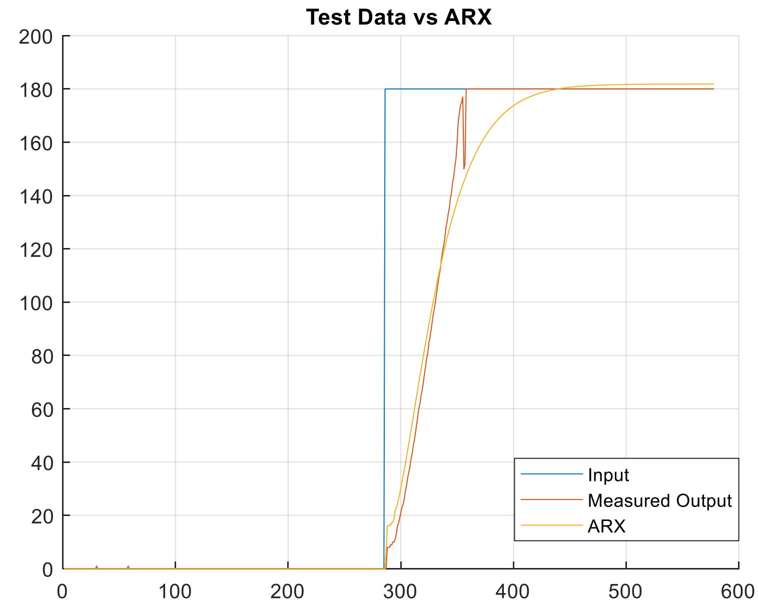
Sinusoid Input:
Best Fit %: 94.505%
FPE: 7.766



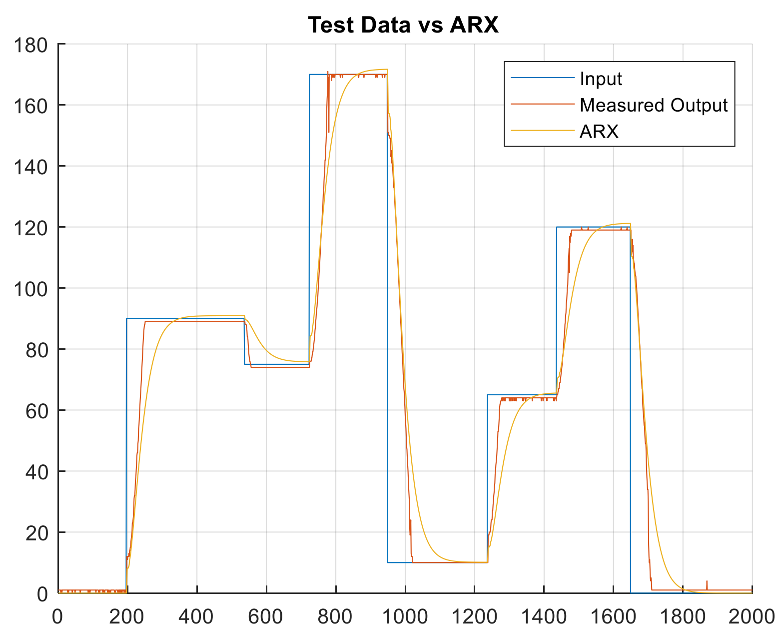
Best Fit %: 94.505%
FPE: 7.7664



Square Wave Input:
Best Fit %: 81.509%
FPE: 5.291



Best Fit %: 81.509%
FPE: 5.2914



Observations:

- For each of the ARX models created, the accuracy of each model was above 80% with each input signal providing slightly different results.
- The models with the best fit also had the largest final prediction error, namely the sinusoidal input to the system.
- Looking at all inputs and metrics, the ordering of each model based on best fit percentage
    1. Sinusoid Input
    2. Step Input
    3. Random Input
    4. Square Wave Input
- Looking at all inputs and metrics, the ordering of each model based on FPE
    1. Step Input
    2. Random Input
    3. Square Wave Input
    4. Sinusoidal Input

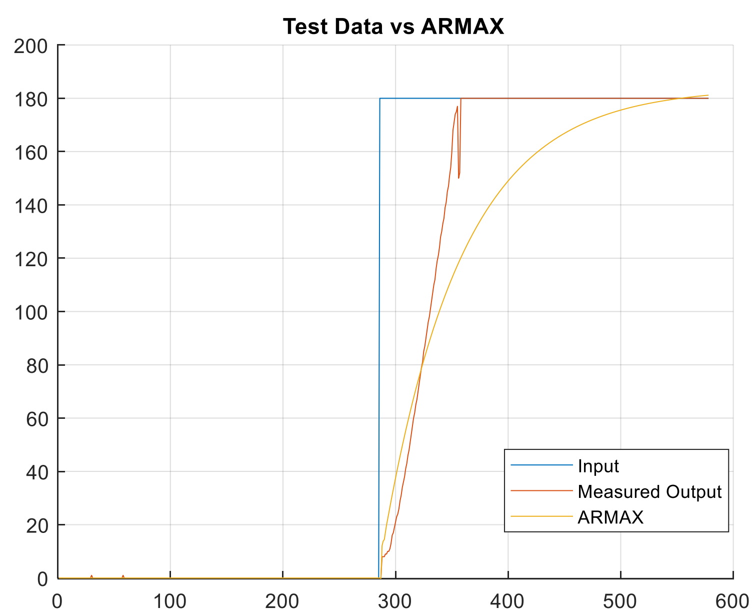
Overall, the sinusoidal input had the highest accuracy, yet the lowest FPE whereas the step input was consistently accurate in its estimation and low in final prediction error. This led to me selecting the model based on the step input as the most effective model.

<u>ARMAX Testing:</u>

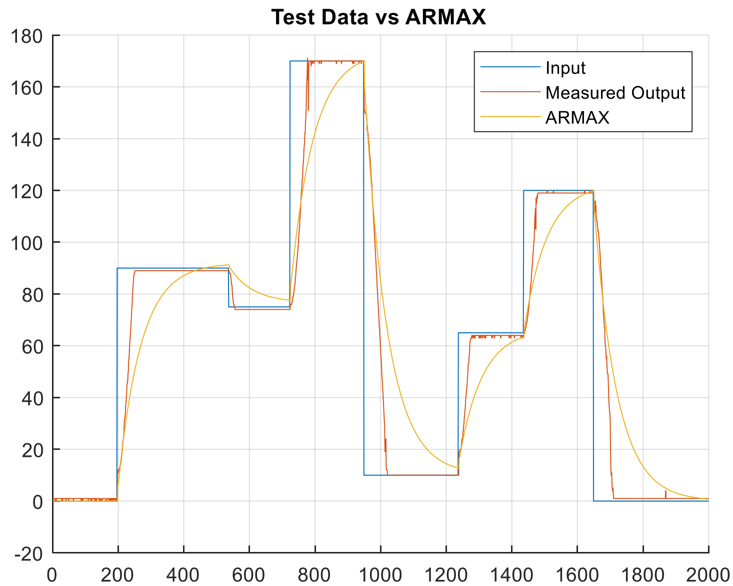
Step Input:

Best Fit %: 90.787%

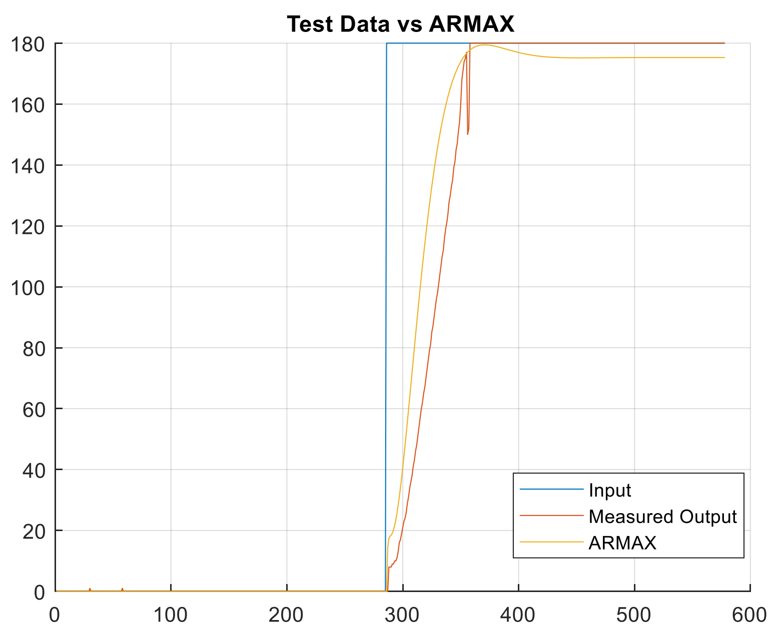
FPE: 1.081



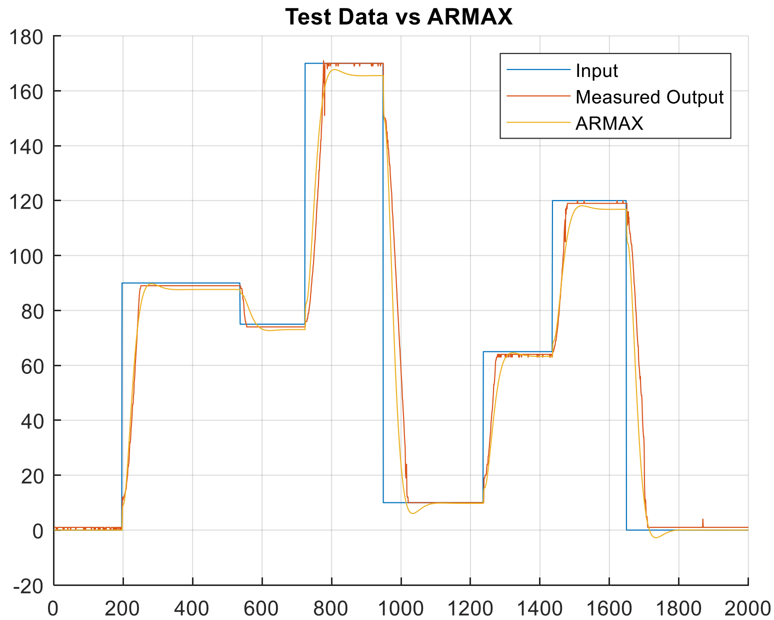
Best Fit %: 90.788%
FPE: 1.0813



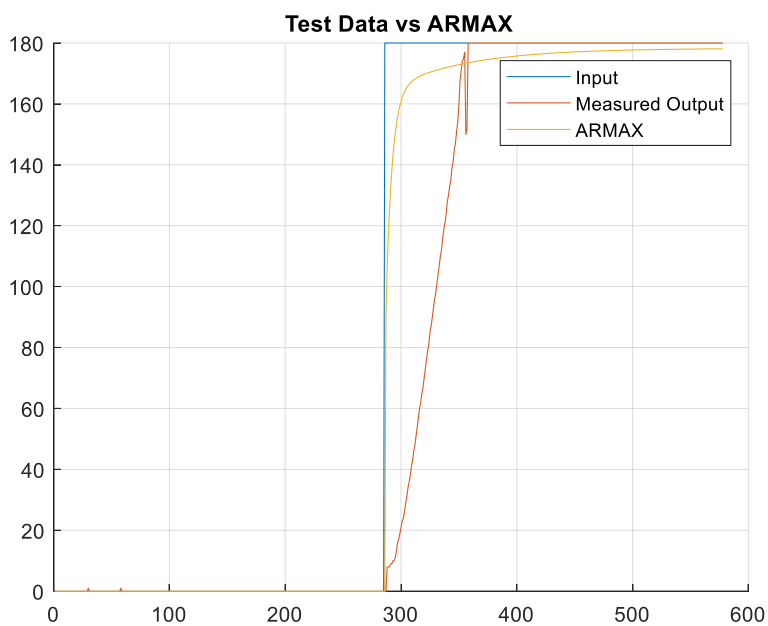
Random Input:
Best Fit %: 87.102%
FPE: 1.879



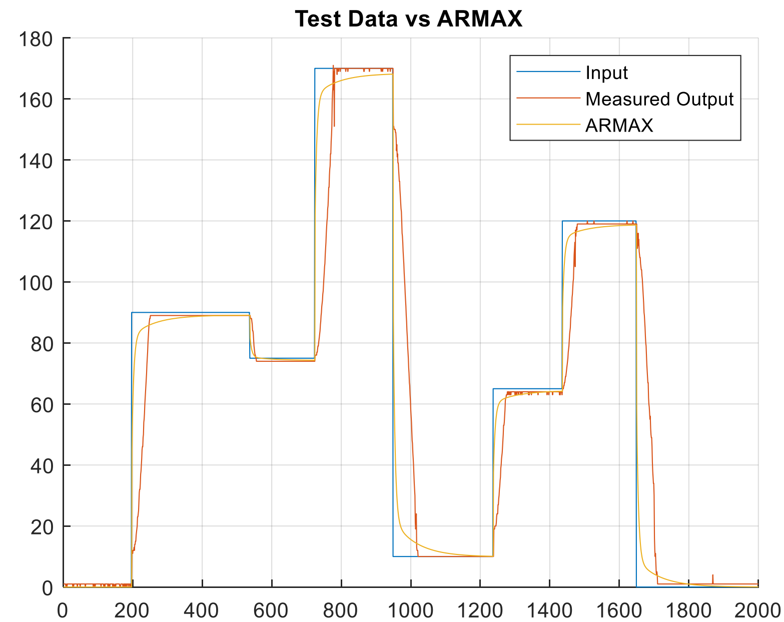
Best Fit %: 87.102%
FPE: 1.879



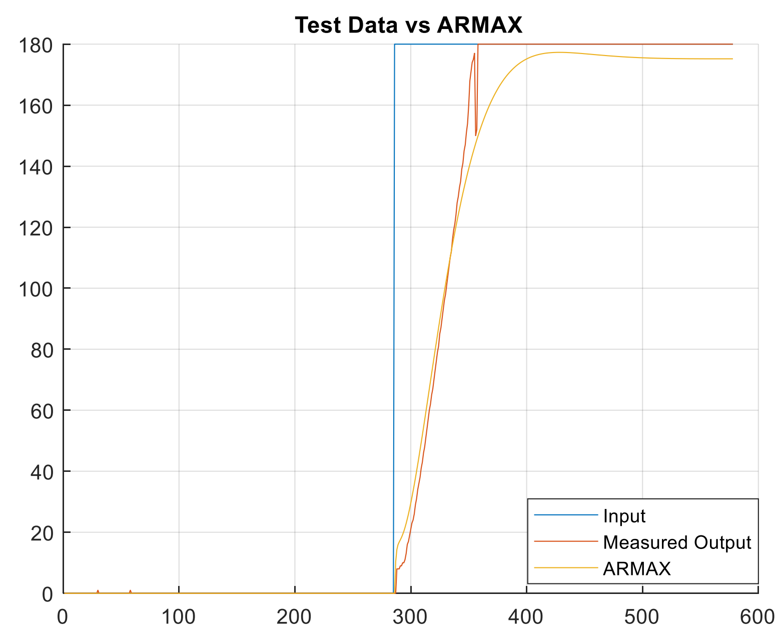
Sinusoid Input:
Best Fit %: 94.956%
FPE: 7.430



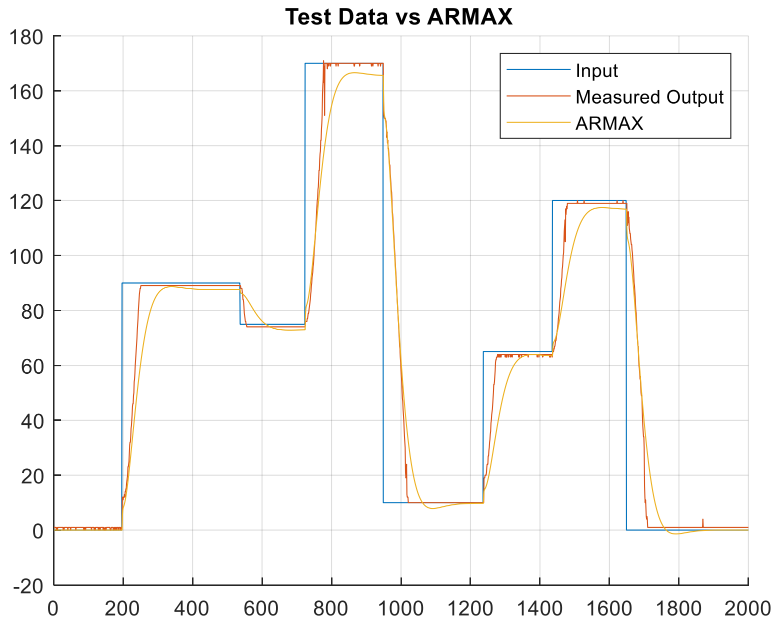
Best Fit %: 94.956%
FPE: 7.430



Square Wave Input:
Best Fit %: 85.862%
FPE: 5.241



Best Fit %: 85.862%
FPE: 5.241



Observations:

- For each of the ARMAX models created, the accuracy of each model was above 80% with each input signal providing slightly different results.
- The models with the best fit also had the largest final prediction error, namely the sinusoidal input to the system.
- Looking at all inputs and metrics, the ordering of each model based on best fit percentage

    5. Sinusoid Input
    6. Step Input
    7. Random Input
    8. Square Wave Input

- Looking at all inputs and metrics, the ordering of each model based on FPE

    5. Step Input
    6. Random Input
    7. Square Wave Input
    8. Sinusoidal Input

APPLICATIONS:

The models generated in this project have good applications in robotics simulations, which have the issues of assuming a linear motion of their motors, or instantaneous motion in some cases. Additionally, these models and techniques can be applied to a slew of other motors and applications of motors to give accurate models of the behavior of these motors for high precision control loops and other simulation applications.

CONCLUSIONS:

Overall, The ARX model performed better than the ARMAX model in my testing, which is the inverse of the results achieved by the paper. However, the ARX model used in this model performed better than the paper which can be attributed to a large number of factors such as the hardware used, number of estimated parameters, estimation methods, etc. The code used in this project exist entirely within the Arduino file ‘621_project.ino’ for the hardware control and the premade input signals as well as ‘FinalProject.m’ for the estimations and graphing.

While investigating the parameter estimations, I found increasing the number of parameters estimate on the ARX model to have a much larger impact on the accuracy of the estimation than in the ARMAX estimation. However, even with the increased number of parameters, I never found the estimation to be overparameterized. This may have been due to my usage of the Systems Identification toolbox instead of using other estimation methods that have been used in this class previously. However, I elected to use the toolbox to better understand the tools provided for the work done in this project. This led to accurate estimations, but with some prediction error when using the periodic inputs. Overall, the usage of the systems identification toolbox was very helpful in this project and led to what I believe to be good estimations of parameters for the system I created.